

MASTER METHODOLOGY • 11-STATE TOMATO GREENHOUSE CONTROL SYSTEM

Complete Scientific Pipeline: Physical Modeling → Soil Substrate Physics → Complete Equilibrium Trim → Unbiased Linearization → Dual Observability → Tri-Hybrid Control → Closed-Loop Diurnal Benchmark

Tomato: *Solanum lycopersicum*

100% Pure Base
MATLAB

Zero Toolbox
Dependencies

Single-File:
main.m

1. SYSTEM INPUTS & HARDWARE SENSORS

$R^{11} \times R^4 \times R^3$

A. 11 Plant State Variables ($x \in R^{11}$):

$$x = [M, T, RH, L, pH, EC, N, P, K, W, VOC]^T$$

M (Moisture %), T (Air Temp °C), RH (Humidity %), L (Light lux), pH (0-14), EC (dS/m), N-P-K (mg/kg), W (Water tank %), VOC (Air Index).

B. 4 Control Actuators ($u \in [0, 1]^4$):

• u_1 : Irrigation Pump | • u_2 : Ventilation Fan

• u_3 : Ultrasonic Mist | • u_4 : Supplemental Grow Light

C. 3 Diurnal Disturbances ($d \in R^3$):

• d_1 : External Temp T_{ext} | • d_2 : Solar Light L_{nat} | • d_3 : Evap E_{vap}

ESP32 Physical Transducer Layer:

BME280 (T/RH via I2C), Capacitive v1.2 (M), BH1750 (L), Analog pH/EC probes, RS485 NPK, Float switch. Injected with realistic Gaussian noise $v \sim N(0, \sigma^2)$.

5. CONTROLLABILITY & DUAL OBSERVABILITY

Kalman Algebraic
Proofs

A. Controllability Matrix & Kalman Staircase:

$$Co = [B, AB, A^2B, \dots, A^{10}B] \in R^{11 \times 44} \Rightarrow \text{rank}(Co) = 4 / 11$$

Mathematical Proof: The 4 environmental climate modes (M, T, RH, L) are fully controllable by the 4 actuators. Unactuated chemical states (5-11) are stabilizable via natural physical dissipation.

B. Dual Observability Analysis (Viva Critical!):

$$\text{Case A (4 Primary Climate Sensors): } C_{\text{prim}} = [I_{4 \times 4}, 0_{4 \times 7}]$$

$$\Rightarrow \text{rank}(Ob_{\text{prim}}) = 4 / 11 \quad \text{Nutrients/Tank Unobservable}$$

$$\text{Case B (Dedicated 11-Transducer Suite): } C_{\text{full}} = I_{11 \times 11}$$

$$\Rightarrow \text{rank}(Ob_{\text{full}}) = 11 / 11 \quad 100\% \text{ Fully Observable}$$

Examiner Takeaway: Air sensors alone cannot reconstruct root-zone NPK or tank reserves. Dedicated chemical probes are mathematically mandatory.

2. NONLINEAR PLANT DYNAMICS & SOIL PHYSICS

Tomato_Dynamics

Coupled System ODEs: $dx/dt = f(x, u, d)$

$$dM/dt = \gamma_{irr} \cdot u_1 - k_{evap} \cdot E \cdot [1 + 0.01 \max(T - 25, 0)] - k_{drain} \max(M - M_{th}, 0)$$

$$dT/dt = 0.15(T_{ext} - T) + 0.00008 L - 3.0 u_2$$

$$dRH/dt = 8.0 u_3 - 4.0 u_2 - 0.15 \max(T - 25, 0)$$

$$dL/dt = 0.5(L_{nat} - L) + 30,000 u_4$$

Transpiration & Fertigation-Leaching Kinetics:

$$\tau = [0.50 + 0.50(L/25000)] \times [1 + 0.02 \max(T - 22, 0)]$$

$$dN/dt = 22 u_1 - 3.8\tau - 0.05 L_{fac} D (N/100) + 0.03(130 - N)$$

$$dW/dt = [0.05(80 - W) + q_{refill}] - (0.5 E + 0.2 u_1)$$

Soil Substrate	Drainage (k_{drain})	Infiltration (γ)	Agronomic Behavior
Sandy	0.045 (Fast)	14.0 (High)	Rapid leaching; requires pulsed fertigation
Loamy (Default)	0.020 (Balanced)	12.0 (Med)	Optimal moisture retention & aeration
Clay	0.008 (Slow)	9.5 (Low)	High water binding; aeration hypoxia risk

6. TRI-HYBRID CONTROLLER SYNTHESIS

Decentralized PID • Pole Placement • Optimal LQR-I

Law 1: Decentralized Multi-Loop PID + Anti-Windup Clamping:

$$u_j(k) = \text{clamp}(K_p e_j + K_i \int e_j dt + K_d de_j/dt, 0, 1)$$

Anti-Windup Logic: Freeze integrator if $(u \geq 1 \wedge e > 0) \vee (u \leq 0 \wedge e < 0)$.

Law 2: Scaled Subspace Pole Placement (Sylvester Solver):

$$(A_{\text{env}} - \lambda_1 I) x_j = B_{\text{env}} g_j \Rightarrow K_{\text{env}} = G X^{-1}, \lambda_{\text{des}} = \{-0.6, -0.8, -1.0, -1.2\}$$

Gain Scaling: Maximum gain $|K_{ij}| \leq 0.2167$ (Eliminates 10^7 gain explosion!).

Law 3: Optimal LQR-I via Real Ordered Schur CARE Solver:

$$\text{Symplectic Hamiltonian } H = [A_a, -B_a R^{-1} B_a^T; -Q_a, -A_a^T] \in R^{30 \times 30}$$

$$[U, T] = \text{schur}(H); \text{ordschur}(U, T, 'lhp'); P = \text{Re}(V_2 V_1^{-1})$$

$$\text{Optimal feedback: } u = u_0^* - K_x \delta y_{\text{meas}} - K_i z_{\text{int}} \bullet \text{Residual} = 1.25 \times 10^{-10}$$

Augmented state $x_a = [x; z] \in R^{15}$ guarantees asymptotic zero steady-state error.

3. COMPLETE 11-STATE EQUILIBRIUM TRIM

Exact Stationary Operating
Point

A. Selected Nominal Environmental Targets:

$$M^* = 60\%, T^* = 25^\circ\text{C}, RH^* = 65\%, L^* = 15,000 \text{ lux}$$

B. Stationary Actuator Equilibrium Inversion:

$$u_1^* = [E_{\text{loss}} + k_{\text{drain}}(M^* - M_{\text{th}})] / \gamma_{\text{irr}} = 15.0\% \text{ (Pump)}$$

$$u_2^* = [0.15(T_{\text{ext}} - T^*) + 0.00008 L^*] / 3.0 = 40.0\% \text{ (Fan)}$$

$$u_3^* = [4.0 u_2^* - (-0.15 \max(T^* - 25, 0))] / 8.0 = 20.0\% \text{ (Mist)}$$

$$u_4^* = \max(0, L^* - L_{\text{nat}}) / 30,000 = 0.0\% \text{ (Light)}$$

C. Consistent Substrate Equilibrium (States 5–11):

$N^* = 132.8 \text{ mg/kg}$, $P^* = 48.2 \text{ mg/kg}$, $K^* = 189.5 \text{ mg/kg}$, $pH^* = 6.37$, $EC^* = 2.41 \text{ dS/m}$, $W^* = 80\%$, $VOC^* = 100$.

$$\text{VERIFICATION GATE: } ||dx_0/dt||_{\infty} = 1.25 \times 10^{-10} \approx 0$$

Guarantees exact mathematical stationary equilibrium across all 11 states simultaneously before linearization!

7. SENSOR LAYER & SAFETY INTERLOCKS

Sensor-in-the-Loop
Architecture

A. Realistic Transducer Layer Simulation:

$$y_{\text{meas}, i} = \text{clamp}(x_i + v_i, y_{\text{min}, i}, y_{\text{max}, i}), v_i \sim N(0, \sigma_i^2)$$

• Temp: $\sigma = 0.20^\circ\text{C}$ • RH: $\sigma = 0.80\%$ • Light: $\sigma = 50 \text{ lux}$ • Moisture: $\sigma = 0.50\%$

Controllers operate strictly on sensed values y_{meas} , not idealized state x .

B. Physical Bounds & Actuator Saturation:

$$x_{\text{bounded}} = \text{clamp}(x, x_{\text{min}}, x_{\text{max}}), 0 \leq M, RH, W \leq 100\%$$

$$\text{Physical Actuator Saturation: } u_{\text{actual}} = \text{clamp}(u_{\text{cmd}}, 0.0, 1.0)$$

C. Automated Water-Level Safety Interlock:

$$\text{IF } x_{10} \text{ (Water Tank)} < 10.0\% \text{ THEN } u_{\text{actual}}(1) = 0.0 \text{ (Pump Cutoff)}$$

Pump Cavitation Protection: Prevents dry-run motor burnout during severe drought or tank drainage faults.

4. UNBIASED LINEARIZATION & STEP PROOF

$A \in R^{11 \times 11}$, $B \in R^{11 \times 4}$

A. Unbiased Numerical Jacobian Differencing:

$$A(:, i) = [f(x_0 + h e_i, u_0, d_0) - f(x_0 - h e_i, u_0, d_0)] / (2h)$$

$$B(:, j) = [f(x_0, u_0 + h e_j, d_0) - f(x_0, u_0, d_0)] / h \quad (h = 10^{-6})$$

Linear state perturbation: $\delta dx/dt = A \delta x + B \delta u$.

B. Open-Loop Eigenvalues & Modal Spectrum:

$$\lambda_M = -0.020 \text{ h}^{-1}, \lambda_T = -0.150 \text{ h}^{-1}, \lambda_L = -0.500 \text{ h}^{-1}$$

$$\lambda_H = 0.000 \text{ h}^{-1} \text{ (Integrator mode)} \bullet \text{Substrate modes } \leq -0.02 \text{ h}^{-1}$$

Step Perturbation Dynamic Consistency Proof:

Step $\delta u_1 = +10\%$ pump for 4h: Linear deviation $\delta M_L = +3.398\%$, Nonlinear deviation

$$\delta M_{NL} = +3.398\%. \text{ Discrepancy: } ||\delta x_{NL} - \delta x_L||_{\infty} < 10^{-9}\%.$$

8. QUANTITATIVE SCORECARD & BENCHMARK

Normalized 24-Hour Multi-
Stage Test

Hierarchical Crop Condition Index (CCI / CESI):

$$CCI = 0.50 \cdot CCI_{\text{env}} + 0.25 \cdot CCI_{\text{soil}} + 0.25 \cdot CCI_{\text{nutr}}$$

Evaluated against FAO-56 & UC Davis physiological growth envelopes.

Performance Metric	Decentralized PID	Pole Placement	Optimal LQR-I
Total ISE (Error Var)	1.53×10^8	8.04×10^8	1.57×10^7 (−89.7%)
Total IAE (Tracking)	50,307.27	118,518.36	5,854.26
Total Variation (TV)	2,090.93	277.93	728.28
Mean Crop Score (CCI)	99.76%	98.37%	99.94%
Steady-State Error	Small offset	Offset	Zero (0.00)

LQR-I Supremacy: Multivariable optimal cross-coordination yields 89.7% error variance reduction and superior noise immunity.

STEP 1
Agro-Plant Model
11 States, Soil Substrates

STEP 2
Equilibrium Trim
 $||dx_0|| = 1.25e-10$

STEP 3
Linearization
Central Diff Jacobians

STEP 4
Observability Test
4/11 vs 11/11 Proof

STEP 5
Tri-Hybrid Control
PID / PP / LQR-I CARE

STEP 6
Sensor & Safety
Noise + Pump Cutoff

STEP 7
24h Benchmark
CCI & Scorecard Export